

Causal Assumptions

Malcolm Barrett

Stanford University

- In *Split Decision: Life Stories*, Ice-T recounts his journey from a life of crime to fame and success
- He contrasts it with the fate of his former crime partner, Spike, now serving 35 years to life
- The book describes how “their paths might have very well been reversed if they had made different choices”

Are they good proxies for each other?

Person	Observed	Missing counterfactual
Ice-T	Abandons criminal life: fame and fortune	Does one more heist: 35 years in prison
Spike	Does one more heist: 35 years in prison	Abandons criminal life: fame and fortune

They differ in more than their choices

- Ice-T's decision was not the only factor in his success: he had enough musical talent to make a career of it
- Spike might differ in hard-to-measure ways, which makes him a poorer proxy than he first seems

We must rely on statistical techniques to help construct these unobservable counterfactuals from observed data

Potential outcomes

- Factual and counterfactual outcomes are two realizations of potential outcomes
- Say the exposure has two levels, $x = \text{continue}$ doing jewel heists or $x = \text{quit}$
- There are two potential outcomes, $y(\text{continue})$ and $y(\text{quit})$, and only one will be realized

The missing potential outcome

- One potential outcome is observable and one is missing; early causal inference methods were framed as missing data problems
- We're missing `y(continue)` for Ice-T and `y(quit)` for Spike, so we can't calculate their individual causal effects

Causal assumptions allow us to imbue observed data with the meaning of potential outcomes and their analysis with causal effects

Does chocolate make you happier?

Suppose some happiness index exists that ranges from 1 to 10

```
1 data <- tibble(  
2   id = 1:10,  
3   y_chocolate = c(4, 4, 6, 5, 6, 5, 6, 7, 5, 6),  
4   y_vanilla = c(1, 3, 4, 5, 5, 6, 8, 6, 3, 5)  
5 ) |>  
6   mutate(causal_effect = y_chocolate - y_vanilla)  
7  
8 data |>  
9   summarize(  
10     avg_chocolate = mean(y_chocolate),  
11     avg_vanilla = mean(y_vanilla),  
12     avg_causal_effect = mean(causal_effect)  
13   )
```

```
# A tibble: 1 × 3  
  avg_chocolate avg_vanilla avg_causal_effect  
      <dbl>         <dbl>         <dbl>  
1          5.4          4.6          0.8
```


Both potential outcomes are known

ID	Potential outcomes		Causal effect
	y(chocolate)	y(vanilla)	y(chocolate) - y(vanilla)
1	4	1	3
2	4	3	1
3	6	4	2
4	5	5	0
5	6	5	1
6	5	6	-1
7	6	8	-2
8	7	6	1
9	5	3	2
10	6	5	1

We can calculate every individual causal effect

In reality, each person eats one flavor

```
1 set.seed(11)
2 data_observed <- data |>
3   mutate(
4     exposure = if_else(
5       rbinom(n(), 1, 0.5) == 1,
6       "chocolate",
7       "vanilla"
8     ),
9     observed_outcome = case_when(
10      exposure == "chocolate" ~ y_chocolate,
11      exposure == "vanilla" ~ y_vanilla
12    )
13  )
14
15 data_observed |>
16   group_by(exposure) |>
17   summarise(avg_outcome = mean(observed_outcome))
```

```
# A tibble: 2 × 2
  exposure avg_outcome
  <chr>      <dbl>
1 chocolate    5.33
2 vanilla      4.71
```

We're a little off because of the low sample size

Only the observed outcome remains

ID	Potential outcomes		Causal effect
	y(chocolate)	y(vanilla)	y(chocolate) - y(vanilla)
1		1	
2		3	
3	6		
4		5	
5		5	
6	5		
7		8	
8		6	
9	5		
10		5	

We can't calculate the individual causal effect

Apples-to-apples

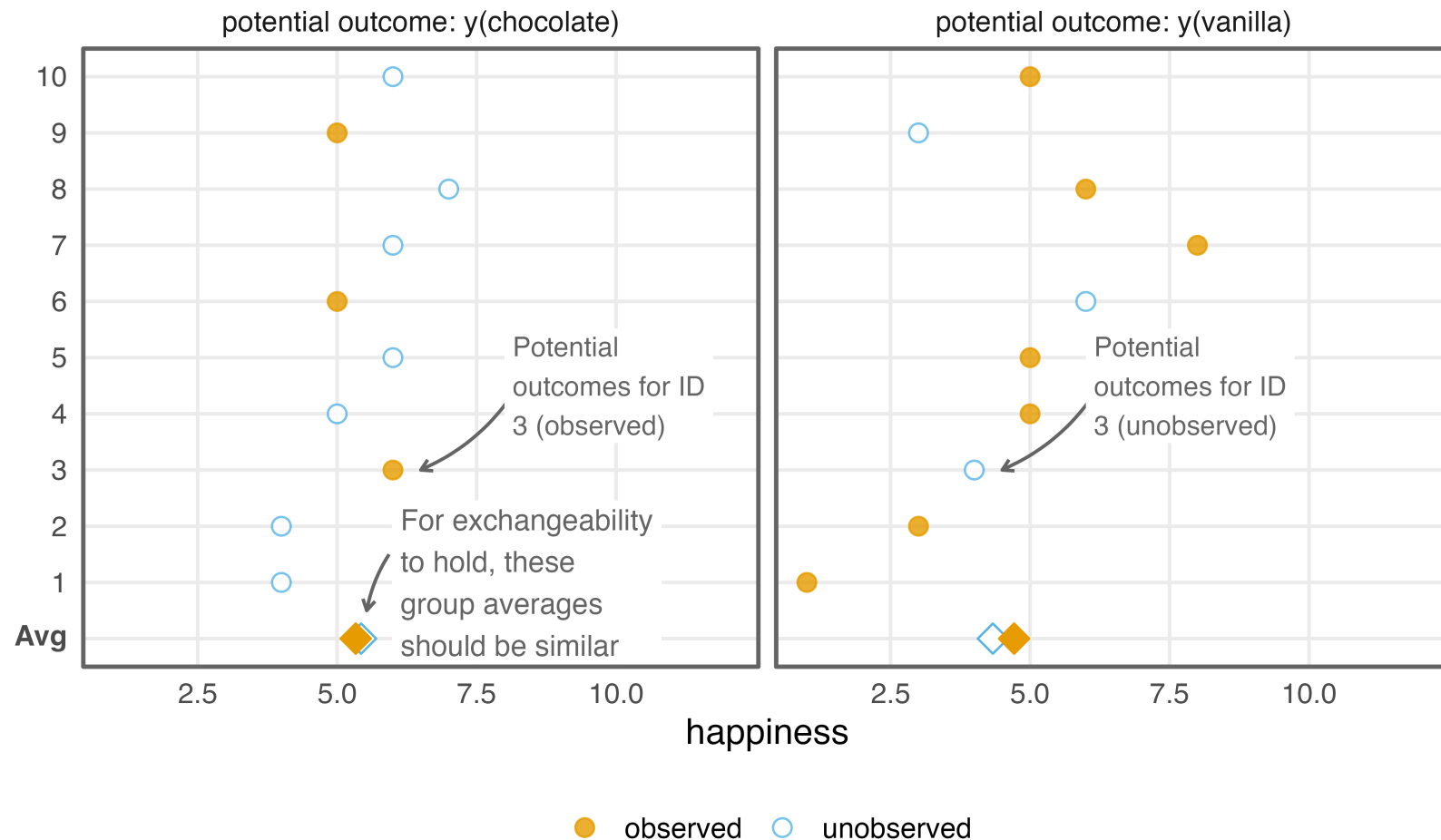
- Most causal assumptions exist to make an apples-to-apples comparison: we want to compare similar individuals who serve as reasonable proxies for each other's counterfactuals
- Other people say cheese and chalk (UK English), apples and pears (German), or grandmothers and toads (Serbian)

The three assumptions

- Unconfoundedness methods assume exchangeability, positivity, and consistency
- Knowing a method's assumptions is essential for using it correctly; these are sometimes called identifiability conditions

Exchangeability

Exposure status is independent of the potential outcomes; that is, each exposure group has the same potential outcomes on average



What if we mixed up the labels?

Say we accidentally gave chocolate to the “vanilla” group and vanilla to the “chocolate” group

```
1 mix_up <- function(flavor) {  
2   if_else(flavor == "chocolate", "vanilla", "chocolate")  
3 }  
4  
5 data_observed <- data_observed |>  
6   mutate(  
7     exposure = mix_up(exposure),  
8     observed_outcome = case_when(  
9       exposure == "chocolate" ~ y_chocolate,  
10      exposure == "vanilla" ~ y_vanilla  
11    )  
12  )  
13  
14 data_observed |>  
15   group_by(exposure) |>  
16   summarise(avg_outcome = mean(observed_outcome))
```

```
# A tibble: 2 × 2  
  exposure avg_outcome  
  <chr>      <dbl>  
1 chocolate    5.43  
2 vanilla      4.33
```

Flavor assignment is independent of the potential outcomes

What if participants chose their own flavor?

```
1 set.seed(113)
2 data_observed_exch <- data |>
3   mutate(
4     prefer_chocolate = y_chocolate > y_vanilla,
5     exposure = case_when(
6       # people choose the flavor they prefer 80% of the time
7       prefer_chocolate ~ if_else(
8         rbinom(n(), 1, 0.8) == 1,
9         "chocolate",
10        "vanilla"
11      ),
12      !prefer_chocolate ~ if_else(
13        rbinom(n(), 1, 0.8) == 1,
14        "vanilla",
15        "chocolate"
16      )
17    ),
18    observed_outcome = case_when(
19      exposure == "chocolate" ~ y_chocolate,
20      exposure == "vanilla" ~ y_vanilla
21    )
22  )
23
24 data_observed_exch |>
25   group_by(exposure) |>
26   summarise(avg_outcome = mean(observed_outcome))
```

```
# A tibble: 2 × 2
  exposure avg_outcome
  <chr>      <dbl>
1 chocolate    5.29
2 vanilla      5.67
```

Now it seems like vanilla makes you happier!

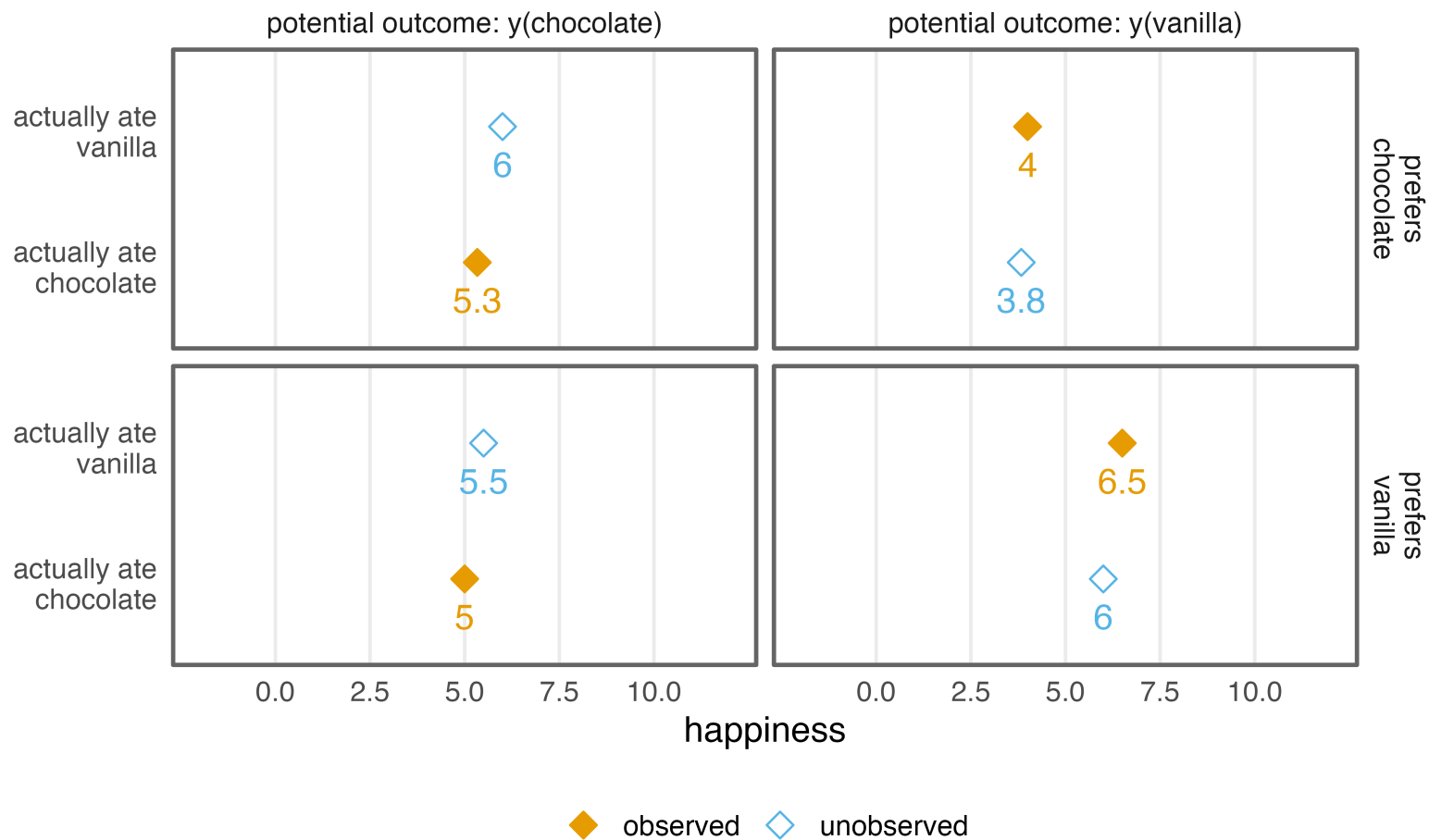
The groups are no longer exchangeable



Preference influences both flavor and happiness

Conditional exchangeability

Exchangeability can hold within preference groups



But few values fall in each combination

Positivity

- Everyone has a non-zero probability of each exposure level
- **Stochastic violations** are chance occurrences with no observations for a given exposure level
- **Structural violations** happen when an individual cannot receive an exposure level by definition

A stochastic violation

Participants choose the flavor they prefer 80% of the time, as before

```
1 data_observed_pos |>
2   count(prefer_chocolate, exposure) |>
3   complete(
4     prefer_chocolate,
5     exposure = c("chocolate", "vanilla"),
6     fill = list(n = 0)
7   )
```

```
# A tibble: 4 × 3
  prefer_chocolate exposure      n
  <lgl>             <chr>    <int>
1 FALSE            chocolate     0
2 FALSE            vanilla       3
3 TRUE             chocolate     7
4 TRUE            vanilla       0
```

Positivity must hold within the covariates needed for exchangeability

A structural violation: an allergy to vanilla

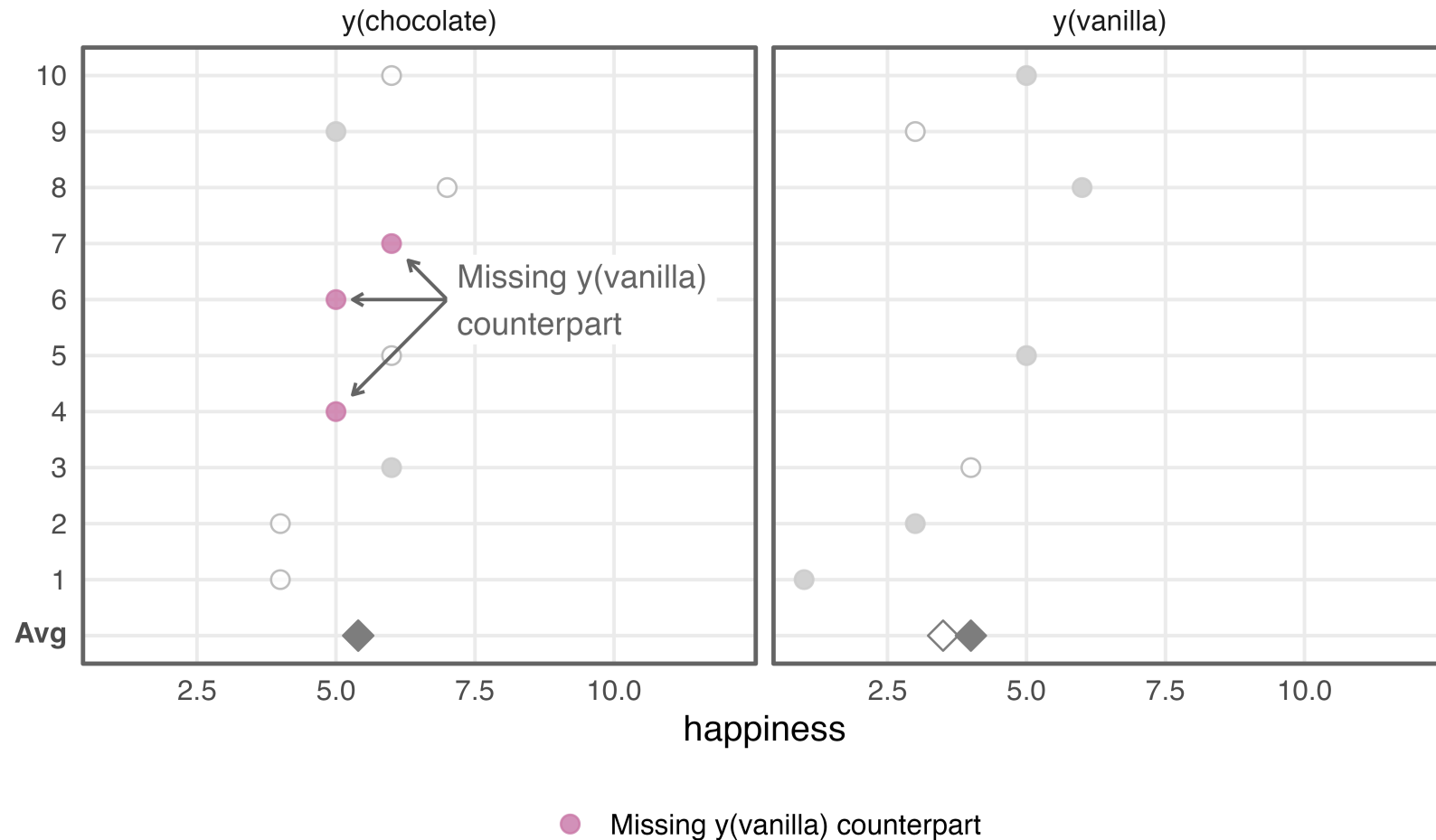
Everyone is randomized to a flavor, as before

```
1 set.seed(1)
2 data_observed_struc <- data_observed_struc |>
3   mutate(
4     allergy = rbinom(n(), 1, 0.3) == 1,
5     # `y_vanilla` is impossible for those with an allergy
6     exposure = if_else(allergy, "chocolate", exposure),
7     y_vanilla = if_else(allergy, NA, y_vanilla),
8     observed_outcome = case_when(
9       allergy ~ y_chocolate,
10      exposure == "chocolate" ~ y_chocolate,
11      exposure == "vanilla" ~ y_vanilla
12    )
13  )
14
15 data_observed_struc |>
16   group_by(exposure) |>
17   summarise(avg_outcome = mean(observed_outcome))
```

```
# A tibble: 2 × 2
  exposure avg_outcome
  <chr>      <dbl>
1 chocolate     5.4
2 vanilla       4
```

For those with a vanilla allergy, `y(vanilla)` is not defined

Undefined potential outcomes



We can collect more data, use a statistical model to extrapolate, add eligibility criteria, or change the estimand

Consistency

- The causal question you claim you are answering is consistent with the one you are actually answering
- The potential outcome of a given treatment value equals the value we actually observe when someone is assigned that treatment value

It seems almost silly when you say it. What else would it be?

Two ways consistency fails

- **Poorly-defined exposure:** multiple treatment versions exist, from surgeries to income to education
- **Interference:** the outcome for any subject depends on another subject's exposure

Two containers of chocolate, one spoiled

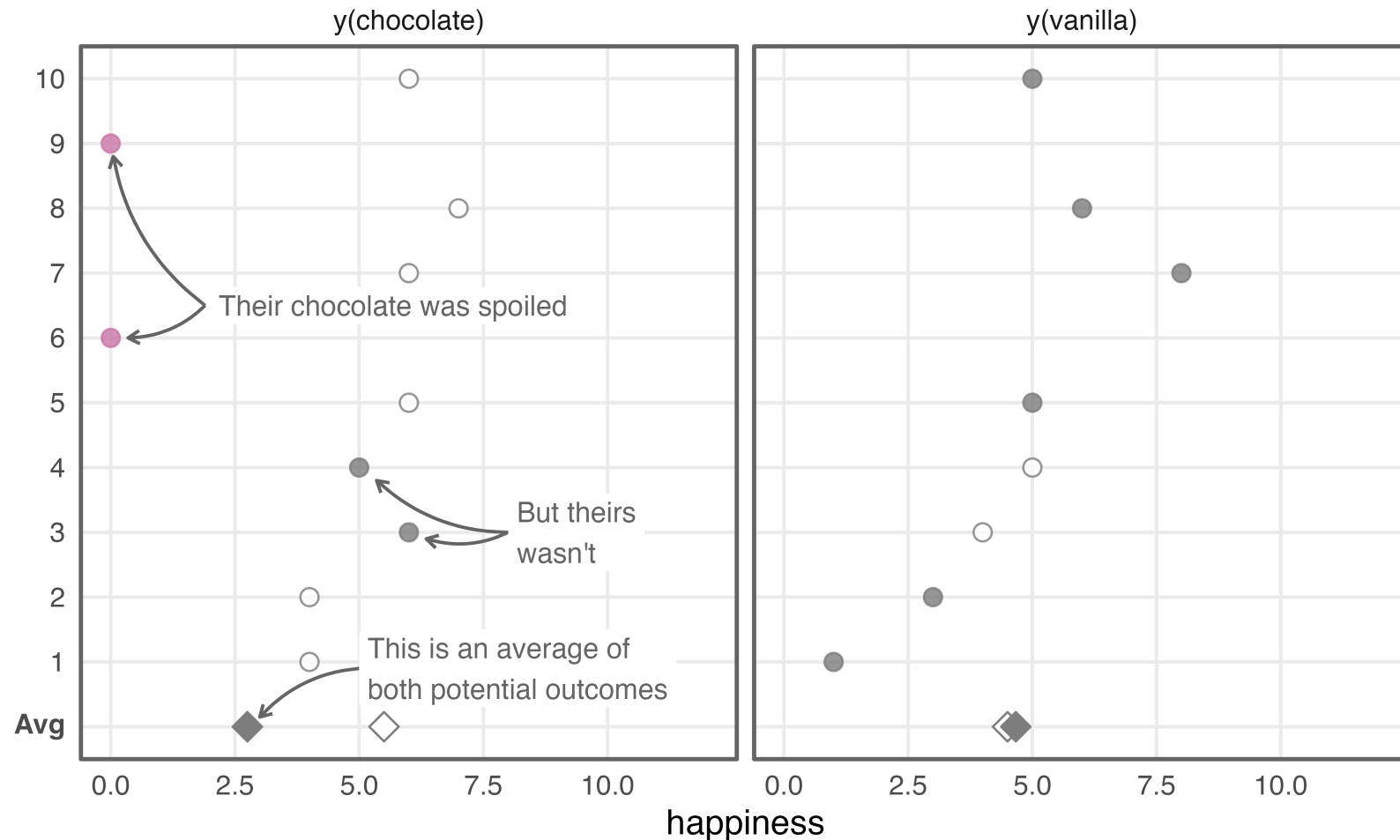
Everyone has a third potential outcome, `y_spoiled_chocolate`, which is 0

```
1 set.seed(11)
2 data_observed_poorly_defined <- data |>
3   mutate(
4     exposure_unobserved = case_when(
5       rbinom(n(), 1, 0.25) == 1 ~ "chocolate (spoiled)",
6       rbinom(n(), 1, 0.25) == 1 ~ "chocolate",
7       .default = "vanilla"
8     ),
9     observed_outcome = case_when(
10      exposure_unobserved == "chocolate (spoiled)" ~ y_spoiled_chocolate,
11      exposure_unobserved == "chocolate" ~ y_chocolate,
12      exposure_unobserved == "vanilla" ~ y_vanilla
13    ),
14    # both containers are recorded as "chocolate"
15    exposure = if_else(exposure_unobserved == "vanilla", "vanilla", "chocolate")
16  )
17
18 data_observed_poorly_defined |>
19   group_by(exposure) |>
20   summarise(avg_outcome = mean(observed_outcome))
```

```
# A tibble: 2 × 2
  exposure avg_outcome
  <chr>      <dbl>
1 chocolate 2.75
2 vanilla   4.67
```

The true causal effect of unspoiled chocolate is 0.8, but our estimate is about -1.9

The chocolate group is a mixture



We're treating fresh and spoiled chocolate as the same exposure; the same goes for brands, timing, and portion size

Separate the potential outcomes

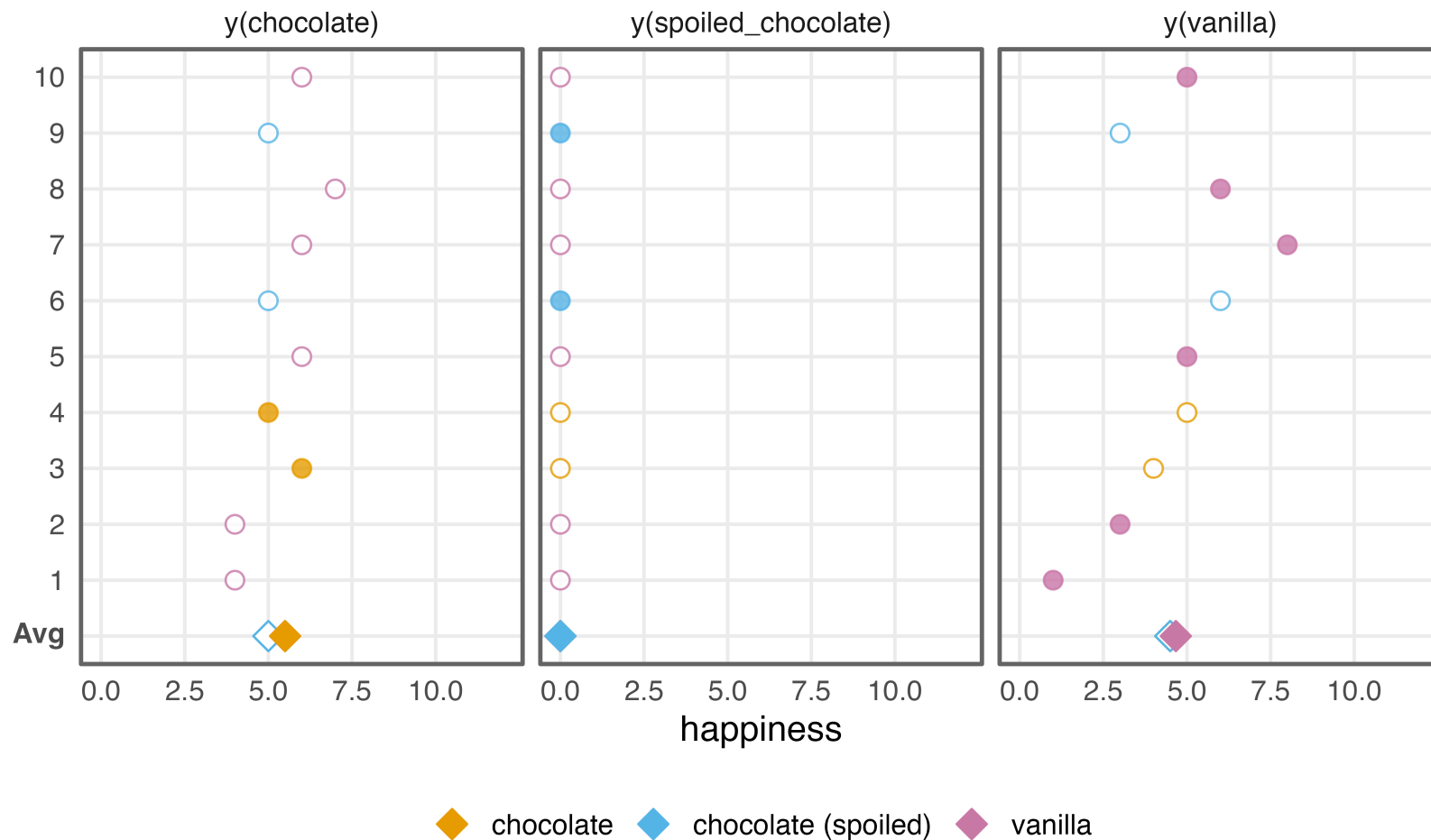
Say we tested the containers after the experiment and know which ones were spoiled

```
1 data_observed_poorly_defined |>  
2   group_by(exposure_unobserved) |>  
3   summarise(avg_outcome = mean(observed_outcome))
```

```
# A tibble: 3 × 2  
  exposure_unobserved avg_outcome  
  <chr>              <dbl>  
1 chocolate          5.5  
2 chocolate (spoiled) 0  
3 vanilla            4.67
```

Now we're back to the right answer because we've correctly separated the potential outcomes

The data match potential outcomes



There will almost always be some consistency violation; the question is whether it is meaningful

Interference

- An individual's exposure impacts another's potential outcome, as when someone's vaccination affects someone else's risk
- Here, each person has a partner, and having a partner with a different flavor increases happiness by two units

Each person has a partner

```
1 data <- tibble(  
2   id = 1:10,  
3   partner_id = c(1, 1, 2, 2, 3, 3, 4, 4, 5, 5),  
4   y_chocolate_chocolate = c(4, 4, 6, 5, 6, 5, 6, 7, 5, 6),  
5   y_vanilla_vanilla = c(1, 3, 4, 5, 5, 6, 8, 6, 3, 5)  
6 ) |>  
7 mutate(  
8   y_chocolate_vanilla = y_chocolate_chocolate + 2,  
9   y_vanilla_chocolate = y_vanilla_vanilla + 2  
10 )
```

Randomize each individual and their partner

```
1 set.seed(37)
2 data_observed_interf <- data |>
3   mutate(
4     exposure = if_else(rbinom(n(), 1, 0.5) == 1, "chocolate", "vanilla"),
5     exposure_partner = if_else(
6       rbinom(n(), 1, 0.5) == 1,
7       "chocolate",
8       "vanilla"
9     ),
10    observed_outcome = case_when(
11      exposure == "chocolate" & exposure_partner == "chocolate" ~
12        y_chocolate_chocolate,
13      exposure == "chocolate" & exposure_partner == "vanilla" ~
14        y_chocolate_vanilla,
15      exposure == "vanilla" & exposure_partner == "chocolate" ~
16        y_vanilla_chocolate,
17      exposure == "vanilla" & exposure_partner == "vanilla" ~ y_vanilla_vanilla
18    )
19 )
20
21 data_observed_interf |>
22   group_by(exposure) |>
23   summarise(avg_outcome = mean(observed_outcome))
```

```
# A tibble: 2 × 2
  exposure avg_outcome
  <chr>      <dbl>
1 chocolate 6.25
2 vanilla   6.33
```

Each person has three counterfactuals now, and these averages don't estimate any of them

Specify the potential outcomes

```
1 data_observed_interf |>
2   group_by(exposure, exposure_partner) |>
3   summarise(avg_outcome = mean(observed_outcome), .groups = "drop")
```

```
# A tibble: 4 × 3
  exposure exposure_partner avg_outcome
  <chr>      <chr>          <dbl>
1 chocolate chocolate        5.5
2 chocolate vanilla          7
3 vanilla    chocolate       6.75
4 vanilla    vanilla         5.5
```

One way to combat interference is to change the unit: randomize the partnerships, a cluster randomized trial

Randomize the partnerships

```
1 set.seed(11)
2 partners <- tibble(
3   partner_id = 1:5,
4   exposure = if_else(rbinom(5, 1, 0.5) == 1, "chocolate", "vanilla")
5 )
6
7 partners_observed <- data |>
8   left_join(partners, by = "partner_id") |>
9   mutate(
10     exposure_partner = exposure,
11     observed_outcome = case_when(
12       exposure == "chocolate" & exposure_partner == "chocolate" ~
13         y_chocolate_chocolate,
14       exposure == "vanilla" & exposure_partner == "vanilla" ~ y_vanilla_vanilla
15     )
16   )
17
18 partners_observed |>
19   group_by(exposure) |>
20   summarise(avg_outcome = mean(observed_outcome))
```

```
# A tibble: 2 × 2
  exposure avg_outcome
  <chr>      <dbl>
1 chocolate     5.5
2 vanilla       4.38
```

There is no interference between different partner sets

Malaria nets, revisited

Malaria nets and the assumptions

- **Exchangeability:** we saw what can happen with an unmeasured confounder, genetic resistance to malaria
- **Positivity:** a violation would occur if any household would always or never use a bed net, or if no one of low economic status in cold weather ever used one

Consistency and interference

- **Consistency:** a “bed net” can mean many things; does the manufacturer matter? The type of insecticide? The amount?
- Interference is a real problem for bed-net research, though households as the unit reduce some of it

When do study designs support causal inference?

Randomized trials

- In the limit, we expect exchangeability:
randomization is the only cause of exposure
- Randomization guarantees positivity so long as no one has a deterministic exposure
- Trials help, but do not guarantee, consistency;
interference can still happen

Trials in practice

- In actual trials, exchangeability can fail by chance, people do not adhere to their assigned exposure, and people drop out
- Observational studies have none of a trial's guarantees, even ideal ones

A randomized design is sometimes called an A/B test

Assumptions solved by study design

Assumption	Ideal randomized trial	Realistic randomized trial	Observational study
Consistency (well-defined exposure)	😊	🙋	🙋
Consistency (no interference)	🙋	🙋	🙋
Positivity	😊	😊	🙋
Exchangeability	😊	🙋	🙋

😊 solved by default, 🙋 solvable but not solved by default

The elements of a target trial

A clear causal question maps to a protocol

- Eligibility criteria: who or what is included?
- Exposure definition: what precise exposure?
- Assignment procedures: how are units assigned?
- Follow-up period: when does it start and end?

More elements of the protocol

- Outcome definition: what precise outcomes will be measured?
- Causal contrast of interest: which causal estimand will be estimated?
- Analysis plan: what procedures will estimate the causal contrast?

How the protocol helps

- Eligibility criteria prevent positivity violations by excluding those who can never have one of the exposures
- A precise exposure definition improves consistency and helps control interference

What randomization adds

- Randomized assignment meets exchangeability and positivity, and can prevent interference

Emulating target trials

- We often cannot randomize: time, money, ethics
- You specify a target trial, the randomized trial you would run, and emulate it with observational data
- The target trial is usually pragmatic: no blinding

This is step 1 of the whole game: specify your causal question, for example as a target trial

Extra Magic Mornings

- Guests staying at a Walt Disney World resort hotel could enter the park during “Extra Magic Hours”
- Does an Extra Magic Morning change the average posted wait time for the Seven Dwarfs Mine Train between 9 AM and 10 AM?

The Seven Dwarfs data

You will work with these data in the propensity score exercises

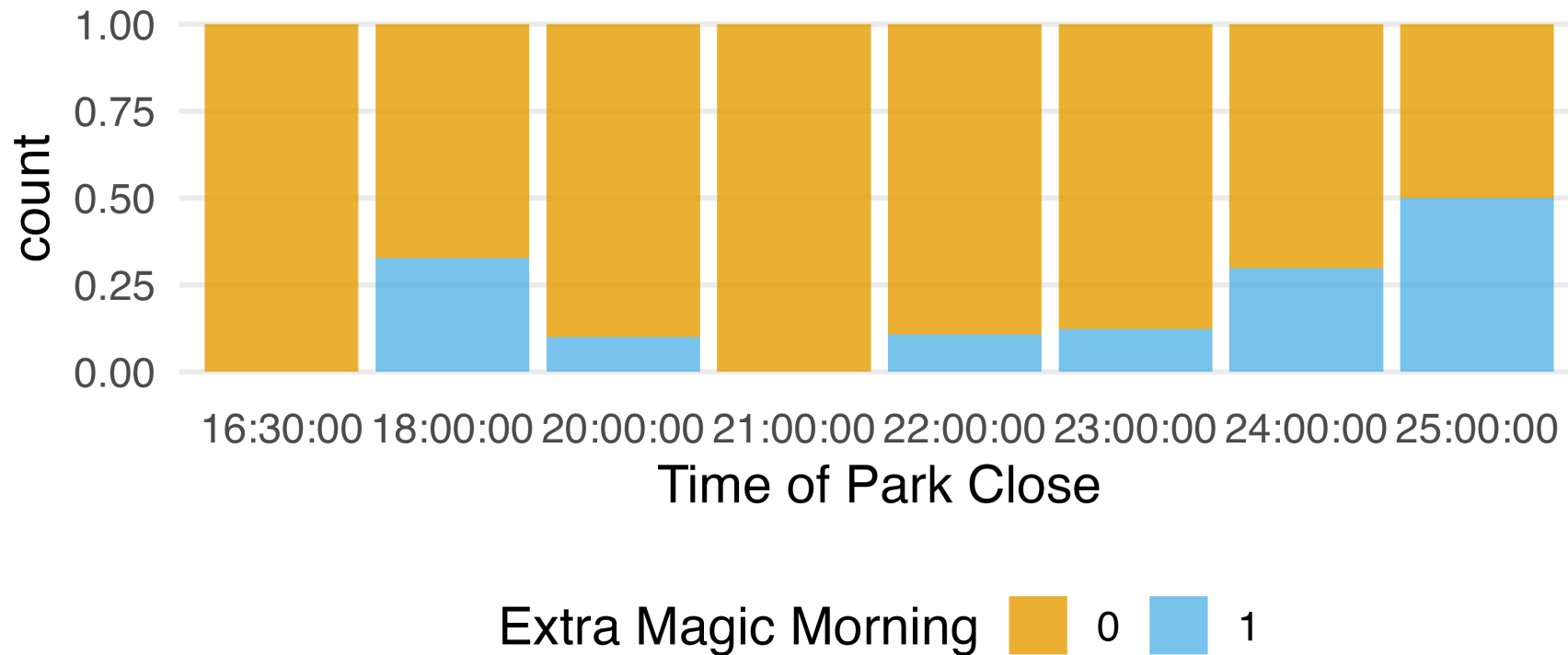
```
1 seven_dwarfs_9 <- seven_dwarfs_train_2018 |>
2   filter(wait_hour == 9)
3
4 seven_dwarfs_9 |>
5   count(park_extra_magic_morning)
```

```
# A tibble: 2 × 2
  park_extra_magic_morning     n
      <dbl> <int>
1             0     294
2             1      60
```

Data can't solve the problem of the unverifiable assumptions, but it can still provide valuable information

Positivity: time of park close

Exchangeability needs better tools, which we'll see later, and we've addressed consistency by being specific about Extra Magic Mornings



None of the days that closed at 16:30 or 21:00 had Extra Magic Mornings; ticket season, by contrast, shows no violations

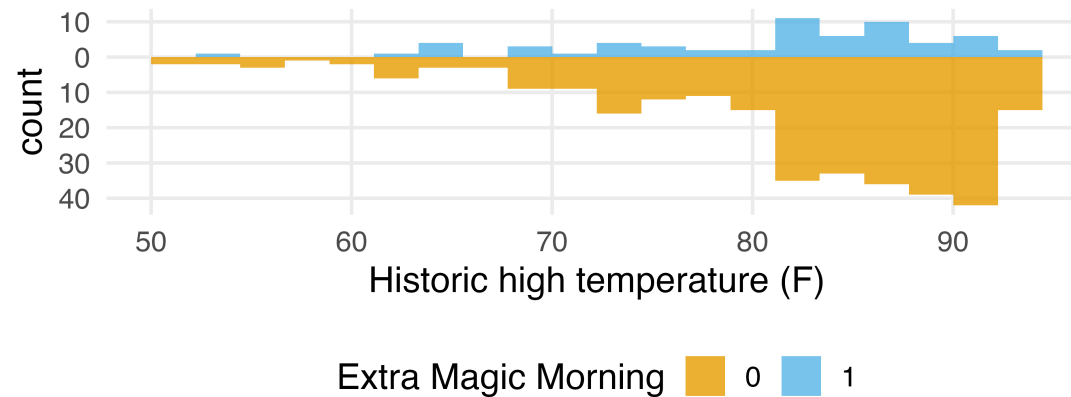
Which days are missing a group?

```
1 seven_dwarfs_9 |>
2   count(park_close, park_extra_magic_morning) |>
3   complete(
4     park_close,
5     park_extra_magic_morning,
6     fill = list(n = 0)
7   ) |>
8   filter(park_close %in% parse_hm(c("16:30", "21:00")))
```

```
# A tibble: 4 × 3
  park_close park_extra_magic_morning     n
  <time>          <dbl> <int>
1 16:30             0         1
2 16:30             1         0
3 21:00             0        28
4 21:00             1         0
```

Only background knowledge can tell us whether a violation is stochastic or structural

Historic high temperature

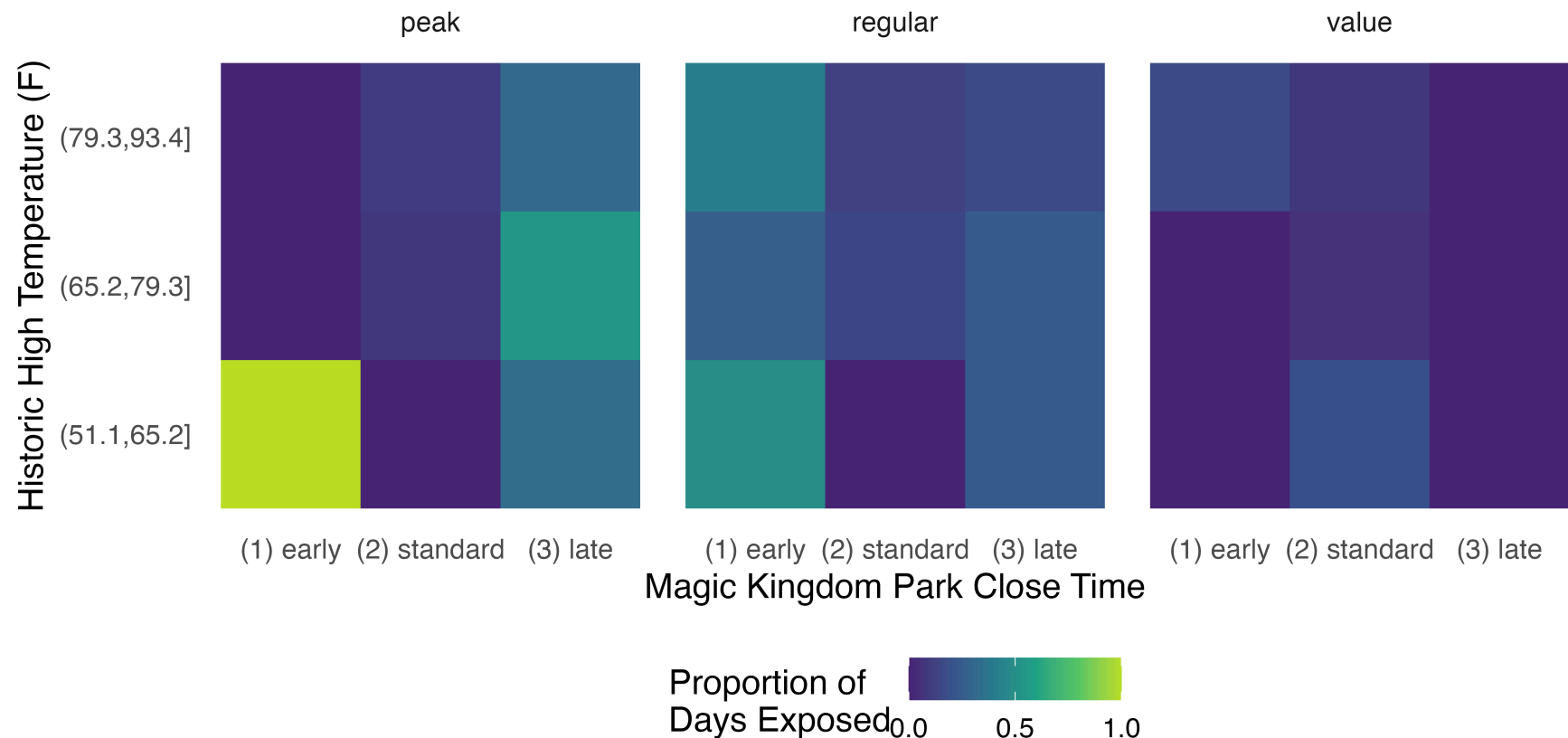


```
1 seven_dwarfs_9 |>
2   filter(park_temperature_high < 60) |>
3   count(park_extra_magic_morning)
```

```
# A tibble: 2 × 2
  park_extra_magic_morning    n
      <dbl> <int>
1             0           9
2             1           1
```

Restricting to warmer days would change the question and restrict which days the conclusions cover

Checking several variables at once

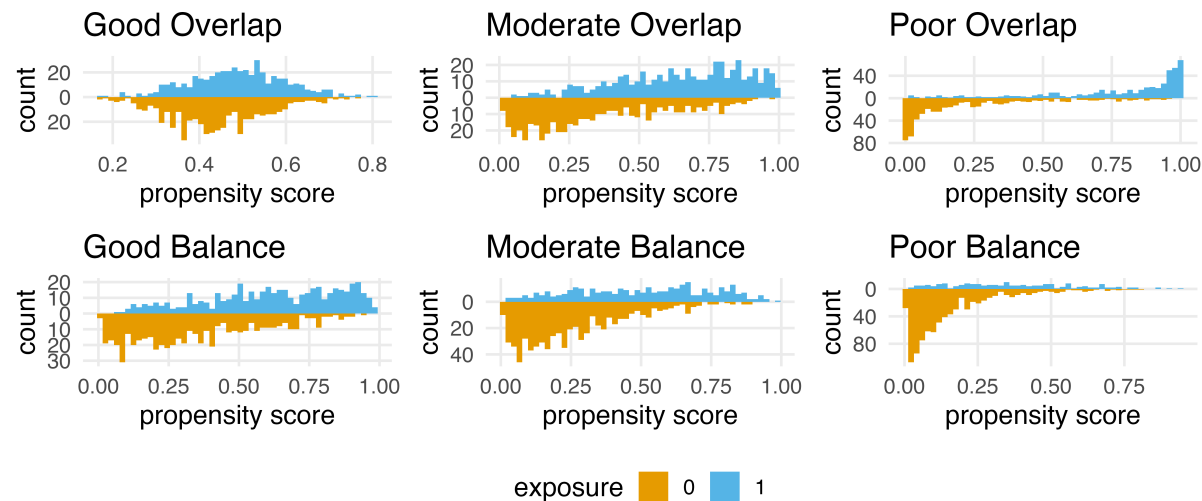


Nine combinations were never exposed, and one was always exposed

The only cold peak days in Florida are over Christmas and New Year, when there historically were always Extra Magic Hours

Overlap and balance

We look for two things: overlap and balance



Overlap, or common support, relates to positivity

Balance relates to exchangeability: we should be able to reassign the two groups and still get the right answer

The x-axis is the probability of exposure given the covariates: the propensity score, which we take up next

Your Turn

Open **07-causal-assumptions-exercises.qmd**

Discuss with your neighbors: what do exchangeability, positivity, and consistency require for the question, does quitting smoking cause weight gain?